

Logistic Regression-Based Classification of Breast Tumors Using the Wisconsin Diagnostic Dataset

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Github: <https://github.com/Rajabali1999/AML-Basic-Project-Breast-Cancer>

Abstract

Breast cancer is one of the leading causes of cancer-related mortality worldwide. Accurate early diagnosis significantly improves patient survival rates. This study presents a binary classification approach for distinguishing benign and malignant breast tumors using Logistic Regression. The Breast Cancer Wisconsin (Diagnostic) Dataset, consisting of 569 samples and 30 nuclear morphological features extracted from Fine Needle Aspiration (FNA) biopsy images, was utilized. A complete machine learning pipeline was implemented, including exploratory data analysis, feature scaling, multicollinearity reduction, baseline comparison using a Dummy Classifier, and logistic regression modeling with class balancing. The proposed model achieved approximately 97% classification accuracy and a Receiver Operating Characteristic Area Under Curve (ROC-AUC) score of 0.99, significantly outperforming the 63% baseline accuracy. The results demonstrate that Logistic Regression, when properly preprocessed, can achieve highly reliable performance in structured biomedical classification tasks.

Keywords— Breast Cancer, Logistic Regression, Binary Classification, Machine Learning, ROC-AUC, Medical Diagnosis.

I. Introduction

Breast cancer remains a major global health concern and a leading cause of mortality among women. Early and accurate tumor classification is critical for determining appropriate treatment strategies. Traditional pathological analysis may involve subjective interpretation, motivating the integration of computational diagnostic support systems.

Machine learning methods provide automated tools for detecting hidden patterns within medical datasets. Logistic Regression is a widely adopted classification algorithm due to its interpretability, computational efficiency, and probabilistic foundation. This study investigates whether Logistic Regression can effectively distinguish malignant from benign tumors using nuclear morphological features.

The main objective is to evaluate the performance of Logistic Regression compared to a baseline classifier while addressing challenges such as class imbalance and multicollinearity.

II. Dataset Description

The dataset consists of 569 biopsy samples described by 30 real-valued features. These features quantify properties of cell nuclei, including:

- Radius
- Texture
- Perimeter
- Area
- Smoothness
- Compactness
- Concavity
- Symmetry

Each characteristic is computed as mean, standard error (SE), and worst value (maximum).

The class distribution is moderately imbalanced:

- 357 benign samples (62.7%)
- 212 malignant samples (37.3%)

The diagnosis variable was encoded as binary:

- Benign = 0
- Malignant = 1

No missing values were observed.

III. Methodology

A. Exploratory Data Analysis

Descriptive statistics revealed substantial variation in feature scales. For example, radius-related features ranged up to 28, while smoothness-related features remained below 0.17. This justified feature standardization prior to model training.

Correlation analysis showed strong multicollinearity among features such as `radius_mean`, `perimeter_mean`, and `area_mean`. Features with correlation coefficients above 0.9 were removed to improve numerical stability and interpretability.

B. Baseline Model

A Dummy Classifier using the “most frequent” strategy was implemented. This classifier predicts all samples as benign and achieved approximately 63% accuracy. This baseline establishes the minimum acceptable performance threshold.

C. Feature Standardization

All features were standardized using Standard Scaler:

$$z = \frac{x - \mu}{\sigma}$$

Scaling parameters were learned from the training set only to prevent data leakage.

D. Logistic Regression

Logistic Regression with L2 regularization was applied. To address class imbalance, the parameter `class_weight='balanced'` was used. The probability of malignancy is modeled as:

$$P(y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta X)}}$$

The dataset was split into 75% training and 25% testing subsets using stratified sampling to maintain class proportions.

IV. Results

A. Classification Performance

The Logistic Regression model achieved:

- Accuracy $\approx 97\%$
- Precision (Malignant) ≈ 0.98
- Recall (Malignant) ≈ 0.94
- F1-score ≈ 0.96
- ROC-AUC ≈ 0.99

Compared to the 63% baseline accuracy, the improvement is substantial.

B. Confusion Matrix Analysis

The confusion matrix indicates:

- 89 benign samples correctly classified
- 50 malignant samples correctly classified
- Only 4 total misclassifications

False negatives (malignant predicted as benign) were minimal, which is critical in medical diagnostics.

C. ROC Curve

The ROC curve approaches the top-left corner of the graph, indicating excellent discriminative ability. An AUC of 0.99 reflects near-perfect class separability.

V. Discussion

The results confirm that nuclear morphological features strongly correlate with tumor malignancy. Logistic Regression performed exceptionally well due to:

1. High-quality structured data
2. Near-linear separability of classes
3. Proper feature scaling
4. Multicollinearity reduction
5. Class weight balancing

Although the performance is strong, limitations include:

- Relatively small dataset size
- Absence of external validation
- Use of engineered features instead of raw imaging data

Future work may include cross-validation, ensemble learning methods, threshold optimization, and dimensionality reduction techniques such as Principal Component Analysis (PCA).

VI. Conclusion

This study demonstrates that Logistic Regression, when combined with appropriate preprocessing and class balancing, can achieve near-clinical performance in breast cancer classification tasks.

The model improved accuracy from 63% (baseline) to 97%, with an AUC of 0.99. These findings highlight the effectiveness of interpretable linear models in biomedical machine learning applications.